



HASTIKA: hate speech and target identification in Kannada-English code-mixed text

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Accepted: 30 April 2025 / Published online: 15 May 2025
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Abstract

In the modern era, the widespread use of social media has facilitated connections among millions of people worldwide. However, these platforms have also been exploited for spreading hate speech, particularly in multilingual contexts. The informal nature of these platforms enables the use of regional languages, leading to code-mixed text. While this linguistic flexibility fosters freedom of expression, it also contributes to the rampant spread of hate speech, posing significant societal challenges. Identifying hate speech in low-resource Kannada-English code-mixed text is challenging due to the scarcity of annotated corpora. To address this gap, the authors introduce HASTIKA (Hate Speech and Target Identification in Kannada-English Code-Mixed Text), a gold-standard corpus specifically designed for hate speech detection and target identification. It consists of 8,058 YouTube comments, annotated for binary classification “Hate” and “Non-hate” and fine-grained categorization into “Gender”, “Political”, “Religion”, “Geo-political”, “Violence”, and “Others”, marking a significant contribution as the first dataset exclusively tailored for this purpose. Topic modeling techniques were applied to uncover latent themes. Benchmark experiments show that fastText, which captures word-level features, achieves 0.7529 accuracy for binary classification and 0.6042 for multi-class classification, while BERT, excelling in sentence-level feature extraction, attains 0.8054 and 0.6819 accuracy, respectively. This research provides a foundational resource for hate speech detection in Kannada-English code-mixed text and underscores the importance of linguistic and contextual information in developing robust classification models.

Keywords Social networks · Hate speech · Code-mixing · Low resource · Corpus building · Kannada-English code-mixed text

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1 Introduction

The widespread adoption of social media has fundamentally reshaped global communication practices. Platforms such as Facebook, Instagram, YouTube, and Twitter have significantly transformed human interactions, information dissemination, self-expression, and knowledge acquisition (Sean, 2022; Zampieri et al., 2019). While these platforms enable rapid and convenient exchange of information, they have also facilitated the proliferation of harmful content, including hate speech, inflammatory language, and misleading representations across the digital landscape. Several factors contribute to the spread of hate speech online, including unchecked freedom of expression, the prevalence of anonymous accounts, inadequate regulatory frameworks for social media, and the existence of echo chambers that reinforce extremist viewpoints. The dissemination of such harmful content, particularly hate speech, poses severe risks to societal harmony and well-being (Schäfer et al., 2021).

Hate speech refers to a form of communication that targets, demeans, or discriminates individuals, groups, or organizations based on characteristics such as gender, race, religion, nationality, ethnicity, sexual orientation, disability, and other attributes (Erjavec & Kovačič, 2012; Biradar et al., 2021; Pamungkas et al., 2020; Banerjee et al., 2021; Mathew et al., 2019). In contrast, offensive language encompasses expressions that may be rude, disrespectful, or inappropriate but do not necessarily incite harm, violence or discrimination (Khan et al., 2023). The fundamental distinction between the two lies in intent and impact-while offensive speech may provoke or insult, hate speech explicitly seeks to incite violence, hatred, or discrimination against a specific group. The consequences of hate speech are far-reaching, leading to the marginalization of specific communities, political instability, psychological harm, and even physical violence. As a result, identifying and mitigating hate speech have become critical areas of focus within the research community (Agarwal & Chowdary, 2021; Biradar et al., 2023; Nascimento et al., 2022; Biradar et al., 2022). A key challenge in addressing hate speech lies in its proliferation within multilingual settings-a global phenomenon where individuals possess proficiency in multiple languages (Agüero-Torales et al., 2021). India, in particular, exemplifies a rich and diverse linguistic landscape, with 22 officially recognized languages and over 700 dialects,¹ India is an ideal testbed for studying the complexities of multilingualism. In such linguistically diverse environments, languages continuously interact and influence one another, resulting in distinctive linguistic phenomena such as code-switching and code-mixing.

Code-mixing, a prominent linguistic phenomenon, refers to the seamless blending of linguistic elements from one language with those of another, creating a distinctive form of communication (Bali et al., 2014; Jose et al., 2020). This process involves the intentional or unintentional integration of vocabulary, syntax, or phonetics from different languages, resulting in a unique communication style. Dravidian

¹ https://en.wikipedia.org/wiki/Languages_of_India.

code-mixing, a phenomenon that occurs when languages such as Kannada, Tamil, Telugu, and Malayalam are mixed with another language, often English, to facilitate expression. Many social media platforms allow users to interact in their preferred languages to enhance engagement and inclusivity, thereby encouraging informal writing practices. Consequently, users often merge their regional or native languages with English or other widely spoken languages to effectively convey their thoughts and perspectives. Winata et al. (2019); Jose et al. (2020).

In a multilingual country like India, studying hate speech across different languages is crucial due to linguistic diversity and cultural variations. Code-mixing is prevalent, making context essential for accurately interpreting tone, intent, social dynamics, and cultural nuances. Context minimizes human errors and reduces the risk of misclassification in automated hate speech detection systems. The identification of hate speech in both monolingual and code-mixed texts, such as Hindi-English, has been a focal point of extensive research (Keya et al., 2023; Kapil et al., 2023; Joshi & Pathak, 2024; Biradar et al., 2022, 2024). Similarly, efforts have been directed toward developing corpora for Dravidian code-mixed languages, including Tamil, Telugu, and Malayalam (Chakravarthi et al., 2020, 2022; Anbukkarasi & Varadhaganapathy, 2023; Hande et al., 2021). However, detecting hate speech in low-resource languages like Kannada-English code-mixed text remains an essential yet underexplored area. Kannada, a Dravidian language and the official language of Karnataka State, India, holds significant cultural and linguistic importance. With approximately 44 million people using Kannada as their primary language and an additional 15 million as their second language,² addressing hate speech in Kannada-English code-mixed text is critical. This linguistic challenge demands focused attention from the research and linguistic communities to ensure inclusivity and mitigate the societal impact of hate speech.

Some of the examples of Kannada-English code-mixed text are as follows:

- “Congress ge mata nedi..Karnataka eega mini pakistanavagalu horatide...congress eega enu maduttade endu nodona!! elliyavaregu congress iratto alliyavar-egu inthaha ghatanegalu nadeyuttiruttave.”

Translation: “Vote for Congress...Karnataka will now become mini Pakistan...Let’s see what Congress will do now!! As long as there is Congress, such incidents will continue.”

- “Ee video maadida uddeshavenu? andare Nehalige muslim janarabagge olavu ittu anta torisabekagide bahusha ivarige.”

Translation: “What is the purpose of this video? It means that Neha had a preference for Muslim people.”

- “Dagar hudgi kachada tara edaale, her parents don’t have brains?”

Translation: “Prostitute girl, she looks like trash. Do her parents not have brains?”

² https://en.wikipedia.org/wiki/List_of_languages_by_total_number_of_speakers.

Handling Kannada-English code-mixed text presents numerous challenges due to its distinct linguistic characteristics. Unlike monolingual text, code-mixed text does not strictly adhere to the grammatical rules of any single language and frequently incorporates mixed lexical and syntactical elements from both languages. One of the primary difficulties in processing such text is accurately identifying the switching points between languages. Additionally, the lack of a systematic and annotated corpus for training machine learning and deep learning models has hindered the development of state-of-the-art solutions for hate speech detection in Kannada-English code-mixed text. To address these challenges, this study focuses on constructing a gold-standard corpus specifically designed for hate speech detection in Kannada-English code-mixed text. The corpus is annotated for binary classification into "Hate" and "Non-hate", with the hate class further categorized into "Gender", "Political", "Religion", "Geo-political", "Violence", and "Others". This work also provides benchmark results as a foundation for future research. To the best of our knowledge, this is the first initiative to develop a hate speech corpus for Kannada-English code-mixed text with fine-grained class annotations. The proposed corpus aims to facilitate the identification of hate speech and its associated targets through binary and multi-class classification frameworks.

The key contributions of the proposed work are as follows:

1. Developed HASTIKA, a novel gold-standard corpus for target-wise hate speech detection in Kannada-English code-mixed text.
2. Proposed a comprehensive pipeline for constructing a corpus in the low-resource Kannada-English code-mixed language setting.
3. Conducted an in-depth topic modeling analysis to identify dominant thematic words within the proposed corpus.
4. Presented baseline results for binary and multi-class classification tasks, establishing a foundation for future research.

The structure of the paper is organized as follows: Sect. 2 reviews existing datasets in Dravidian languages and the methodologies employed to process them. Sections 3 and 4 outline the label taxonomy and the process of raw data collection, respectively. Section 5 details the annotation process, including human annotation, annotator selection, guidelines, data labeling procedures, and inter-annotator agreement. Topic modeling is discussed in Sect. 6, followed by an in-depth corpus analysis in Sect. 7. Section 8 describes the baseline methods and in-context learning approaches utilized in this study. The results and discussion of these baseline methods are presented in Sect. 9. Finally, Sects. 10 and 11 address the limitations of the proposed work and provide conclusions along with directions for future research.

2 Related work

Social media data is a valuable resource for various natural language processing (NLP) tasks, including language identification, sentiment analysis, offensive content detection, and hate speech identification. Its diverse forms of expression and real-time nature provide rich contextual information, though they also introduce challenges stemming from linguistic complexity and diversity. While English dominates most social media platforms, resulting in the availability of substantial annotated corpora for numerous NLP tasks, the increasing accessibility of the internet and smartphones has brought a more diverse demographic to these platforms. This has led to the emergence of code-mixed text, where users blend their native languages with English to express their thoughts. Despite the growing prevalence of code-mixed text, annotated corpora for such data remain scarce due to the linguistic diversity and complexity involved. Recently, there has been a surge of interest among researchers in addressing the challenges associated with processing code-mixed text. Given the limited availability of datasets for Kannada-English code-mixed text in low-resource settings, this study leverages existing corpora developed for various NLP tasks in Dravidian languages as part of the literature review. The Dravidian language family—which includes Tamil, Telugu, Malayalam, and Kannada—exhibits similar phonetic and linguistic structures due to their shared ancestry, making it a valuable reference point for this research.

The literature review is organized into two sections:

1. An overview of the available corpora in Dravidian languages for various tasks, including language identification, sentiment analysis, offensive content detection, hate speech identification, and hate speech.
2. A review of the methodologies used for processing Dravidian languages to identify offensive and hate speech.

2.1 Corpus available in Dravidian languages

Table 1 provides a summary of several existing corpora. Lakshmi and Shambhavi (2017) developed a corpus for language identification in code-mixed data sourced from Facebook and Twitter. This dataset includes over 6,250 words in Kannada, 6,000 words in English, and 500 code-mixed words at the word level. The authors applied various supervised classification techniques for language identification, with the dictionary-based approach achieving the best performance in recognizing word-level code-mixing with an accuracy of 94.8%. Similarly, Thara and Poornachandran (2021) created a corpus of code-mixed Malayalam-English data, focusing on word-level language identification (WLLI). The data was sourced from YouTube, and labeled with Malayalam, English, acronyms, universal, mixed, and undefined categories. Among the pre-trained deep learning models employed for word-level language detection, ELECTRA showed highest accuracy of 0.99.

Table 1 List of corpus available in Dravidian languages

Article	Task	Language	Year	Labels	Size
Lakshmi and Shambhavi (2017)	Language Identification	Kanglish	2017	–	6250 Kannada words 6000 English 500 Code-mixed 50k sentences
Thara and Poornachandran (2021)	Language Identification	Manglish	2021	–	
Kannadaguli (2021)	Sentiment Analysis	Kannada-english	2021	Constructive, Destructive, Assorted, and Undecided	15,398
Chakravarthi et al. (2020)	Sentiment Analysis	Tamil-English	2020	Positive, Negative, Mixed, Neutral, Other language	15,744
Chakravarthi et al. (2020)	Sentiment Analysis	Malayalam-English	2020	Positive, Negative, Mixed, Neutral, Other language	6739
Chakravarthi et al. (2022)	Sentiment Analysis and Offensive language identification	Tamil-English Kannada-English Malayalam-English	2022	Positive, Negative, Mixed, Neutral, Other language	Tamil-42,133 Malayalam- 18,403 Kannada-5,874
Hande et al. (2020)	Sentiment Analysis and Offensive language identification	Kannada-English, Kannada	2020	Positive, Negative, Mixed, Neutral, Other language	7761

Table 1 (continued)

Article	Task	Language	Year	Labels	Size
Hande et al. (2021)	Offensive language identification	Tamil-English Kannada-English Malayalam-English	2021	Offensive Targeted Individual, offensive Targeted Group, offensive Targeted others Not in intended Language	Tamil- 43,909 Malayalam-20,010 Kannada-7,772
Anbukkarasi and Varadhaganapathy (2023)	Hate speech detection	Tamil-English	2023	Hate, Non-hate	10,000
Chakravarthi (2020)	Hope speech detection	Kannada-English	2021	Hope, Non-hope speech	6176
Khanduja et al. (2024)	Hate speech detection	Telugu	2024	Negative, Positive	38,035

Kannadaguli (2021) developed a Kannada-English code-mixed corpus for sentiment analysis, comprising 15,398 comments collected from social media platforms such as Twitter, Facebook, and YouTube. The dataset is labeled into four categories: constructive, destructive, assorted, and undecided. To evaluate the dataset, widely recognized machine learning algorithms were employed using the TF-IDF feature vectors. In addition, deep learning models such as BERT, Bi-LSTM, and Contextualized Dynamic Meta Embeddings (CDME) were implemented. Decision Trees(DT), Logistic Regression(LR), and Random Forest(RF) shows substantial precision, recall, and F1-scores and CDME consistently demonstrated strong performance across all classes. Chakravarthi et al. (2020) created a sentiment analysis dataset for Tamil-English code-mixed text using comments scraped from YouTube. This dataset, which contains 15,744 comments, is labeled with categories such as positive, negative, mixed feelings, neutral, and other languages. The authors employed various machine learning algorithms, along with deep learning models such as 1D Convolutional LSTM (1DConv-LSTM), BERT-Multilingual, Dynamic Meta Embedding (DME), and Contextualized Dynamic Meta Embedding (CDME) to establish benchmark results. Among these models, LR, RF demonstrated superior performance across all sentiment classes, achieving F-scores of 0.68. Similarly, Chakravarthi et al. (2020) developed a Malayalam-English sentiment analysis dataset, also sourced from YouTube comments, and categorized it into five distinct classes: Positive state, Negative state, Mixed feelings, Neutral state, Not in intended language. This dataset includes 6,739 comments. To establish benchmark results for the proposed dataset, the same set of machine learning and deep learning models as utilized in Chakravarthi et al. (2020) were implemented. With the exception of Support Vector Machines (SVM), all ML models successfully classified all sentiment classes. Additionally, the 1D Convolutional LSTM (1DConv) model achieved the highest macro-averaged score across the evaluation metrics.

In Chakravarthi et al. (2022), a corpus was constructed for identifying offensive content and performing sentiment analysis across three major Dravidian code-mixed languages: Tamil-English, Kannada-English, and Malayalam-English. The authors collected data from movie trailers to detect offensive content and perform sentiment analysis, resulting in a data set that contains 66,410 total comments. In establishing the benchmark results, the machine learning algorithms performed better across all sentiment classes and highest F-score 0.59 was achieved by BERT, XLM, and Char-BERT. LR and RF exhibited relative better performance in offensive content detection, while the highest weighted average F-score of 0.74 was observed from Distil-BERT. Similarly, Hande et al. (2020) developed a dataset for Kannada- English and Kannada languages, consisting of 7,671 YouTube comments. The data was manually annotated by at least three annotators for each comment and released for multi-task learning, with the dataset labeled into five categories. State-of-the-art machine learning models were employed to establish baseline results. Logistic Regression model demonstrated superior performance with an accuracy of 0.59 for sentiment classification 0.66 for offensive language identification. Additionally, Hande et al. (2021) created a dataset for offensive content detection in code-mixed Dravidian languages, including Tamil-English, Kannada-English, and Malayalam-English. This dataset comprises 71,691 comments, all transliterated into their respective

Dravidian scripts, with pseudo-labels generated for the transliterated dataset. The original code-mixed comments, along with their transliterated counterparts and pseudo-labels, were combined to create a custom dataset named CM-TRA. Several pre-trained models, including mBERT, XLM-RoBERTa, IndicBERT, DistilBERT, MuRIL, and ULMFiT, were fine-tuned on this newly constructed dataset. Among these, the fine-tuned ULMFiT model demonstrated superior performance with F-score of 0.96. Anbukkarasi and Varadhaganapathy (2023) developed a dataset for hate speech identification in Tamil-English code-mixed text, sourced from Twitter. The dataset comprises 5,000 instances each of hate and non-hate speech, annotated for binary classification. The authors employed recurrent neural network models, including LSTM, Bi-LSTM, and 1DConv, for the classification task. Among these, the Bi-LSTM model demonstrated superior performance with accuracy of 0.82 classifying both categories. Chakravarthi (2020) constructed a dataset for detecting Hope Speech, which includes 6,176 user-generated comments in code-mixed Kannada scraped from YouTube. Each comment was manually labeled as either Hope Speech or Non-Hope Speech. The authors introduced the DC-BERT4Hope model, which leverages the English translation of the Kan-Hope dataset for Hope Speech detection. This approach achieved a weighted F1-score of 0.756.

2.2 Methods used for Dravidian languages

Researchers have shown considerable interest in identifying and mitigating the spread of hate speech. While significant progress has been made in high-resource languages like English, exploring low-resource code-mixed texts, such as Kannada-English, remains limited. Nevertheless, as highlighted in this review, researchers have made notable contributions in identifying hate and offensive content in low-resource Dravidian languages.

Vasantharajan and Thayasivam (2022) conducted extensive experiments using various deep learning and transfer learning models to identify hateful and offensive content on YouTube. They introduced an ensemble multilingual transformer network combining BERT, DistilBERT, and XLM-RoBERTa. They employed Universal Language Model Fine-Tuning (ULMFiT) alongside mBERT and BiLSTM to classify offensive content in Tamil code-mixed data, with ULMFiT achieving highest F-score of 0.74. Subramanian et al. (2022) examined the HASOC offensive language identification track data set in the FIRE2021 Dravidian code mix data set. Initially, they employed traditional machine learning models to detect offensive text, including Support Vector Machine, Logistic Regression, K-Nearest Neighbor, and Bernoulli Naïve Bayes. Subsequently, they experimented with NLP-based pre-trained multilingual transformer models, such as MuRIL, mBERT, and XLM-RoBERTa, utilizing fine-tuning and adapter transformers. The results revealed that the adapter-based XLM-RoBERTa model achieved the highest accuracy of 88.5%. Similarly, Balakrishnan et al. (2023) compared supervised and unsupervised learning methods for detecting offensive language in Tamil. They applied K-means clustering to group unlabeled data before training machine learning algorithms for offensive language detection. Notably, when combined with ensemble classifiers,

the unsupervised clustering method outperformed models trained on human-labeled data with an impressive accuracy of 99.7% and 99.87% for balanced and imbalanced datasets, respectively. Vadakkekara Suresh et al. (2021) employed Model-Agnostic Meta-Learning (MAML) and ProtoMAML to detect offensive content in low-resource Tamil-English and Malayalam-English code-mixed texts. Initially, XLM-RoBERTa was trained using these meta-learning algorithms on code-mixed and monolingual data. The model was then fine-tuned for the target tasks to identify offensive content. Experimental results demonstrated that metalearning significantly improved performance in low-resource, few-shot learning tasks with an F-score of 0.82.

Several shared tasks have been organized to advance research on offensive content detection in low-resource Dravidian code-mixed languages. These include machine translation for Dravidian languages (Kumar et al., 2019), abusive comment detection (Priyadharshini et al., 2022), offensive span identification in Tamil-English code-mixed comments (Ravikiran et al., 2022), conversational hate speech detection in the HASOC-2021 code-mixed language track (Satapara et al., 2021), and offensive language identification in Dravidian code-mixed text (HASOC FIRE 2020) for Tamil, Malayalam, and Kannada (Chakravarthi et al., 2020, 2021). The organization of these shared tasks has garnered significant attention from the research community. For instance, Yasaswini et al. (2021) employed a range of transformer-based models, including CNN-BiLSTM, mBERT, DistilBERT, XLM-RoBERTa, and ALBERT, to classify text in Malayalam, Tamil, and Kannada. Biradar and Saumya (2022) applied multilingual BERT for feature extraction, followed by various classifiers to categorize content as offensive or non-offensive. Additionally, Palanikumar et al. (2022) enhanced the dataset by transliterating it using the ai4bharat Python package and augmenting the dataset size. They employed CountVectorizer for text vectorization and a TF-IDF transformer for feature transformation, followed by machine learning models such as LightGBM, CatBoost, Random Forest, and Support Vector Machines to classify abusive text. In another study, Biradar and Saumya (2022) utilized the pre-trained IndicBERT model for feature extraction, coupled with a Support Vector Machine classifier for stance detection, achieving an accuracy of 62% on Tamil-English code-mixed text. These studies demonstrate the active participation and diverse methodologies employed to address the challenges of offensive content detection in low-resource Dravidian code-mixed languages.

3 Label taxonomy

The dataset undergoes a two-tier annotation process, as illustrated in Fig. 1. In the first stage, annotators classify each comment as either “Hate” or “Non-hate”. In the second stage, hate speech comments are further categorized into specific subcategories based on the content and intent expressed in the text. These subcategories include “Gender”, “Political”, “Religion”, “Geo-political”, “Violence”, and “Others”. This fine-grained classification provides a deeper understanding of the various manifestations of hate speech, offering valuable insights for developing targeted

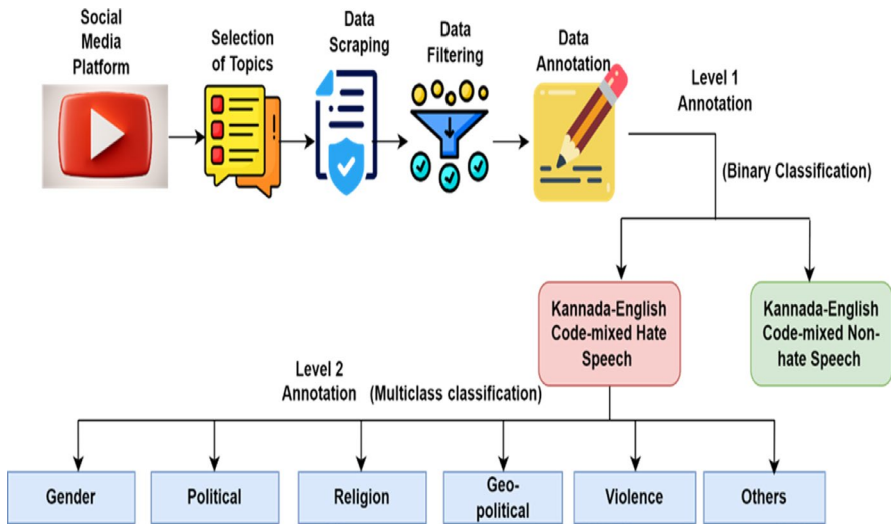


Fig. 1 HASTIKA annotation pipeline

interventions aimed at mitigating harmful content and promoting more supportive discourse.

The taxonomy of the labels used for annotation is as follows:

3.1 Binary class labels

Initially, the problem was approached as a binary classification task with the following labels:

- **Hate:** Hateful comments encompass pejorative language, prejudice, animosity, and intolerance directed towards individuals, groups, or organizations based on attributes such as gender, religion, nationality, disability, sexual orientation, or other protected characteristics. Comments that fall under this category are labeled as “Hate”.

Example 1 “Modlu aga adhikara idda thukali sarkarana chappalili hodibeku boli makianna bhikshukaranna madbeku”

Translation: “First, we need to beat this worthless government with slippers and turn these fools into beggars.”

Label: Hate

- **Non-hate:** This category includes comments that do not contain derogatory language or exhibit animosity towards individuals or organizations. These comments may consist of opinions, constructive criticism, or informational content. Comments in the non-hate class are labeled as “Non-hate”.

Example 2 “Power Star Appu fan here, but Darshan sir neev great eethara stage meleheloke guts beku, kannada bagge nimge thumba gourava ide..”

Translation: “I’m a fan of Power Star Appu, but Darshan sir, you’re great. It takes guts to speak like this on stage. You have a lot of respect for Kannada”

Label: Non-hate

3.2 Multi class labels

Subsequently, each hate label is further classified into more fine-grained categories including:

- **Gender:** Comments using abusive language showing hostility, bigotry, or prejudice towards any individual based on their gender, such as men, women, and transgender people. Such comments have been designated with a label “Gender”.

Example 3 “Ivramma mostly Majestic under pass client irbeku, illaandree etara prathabivantha hennumaklige janmasigodu kasta”

- **Translation:** “His mother must be a Majestic Underpass client, otherwise it is difficult to give birth to such a talented girl.?”

Label: Gender

- **Political:** Comments that consists of statements expressing hatred towards political parties, political leaders, issues, and ideologies. Comments in this category are assigned with the label “Political”.

Example 4 “Nayakare vandematharam heloke asahyapadthiro kanthri samskara illada congresgrinda yenu nirikshisabahudu sir”

- **Translation:** “Leaders feel saying VandeMataram is disgusting, what can you expect from the Congressmen, sir.”

Label: Political

- **Religion:** Comments that indicate hatred for any specific religion, religious beliefs, practices, priesthood, or caste are classified as religion-based hate speech. These comments are assigned with label “Religion”.

Example 5 “Hindhugalu devarunambhali nambhade irale adre avastokke avarithare adaree muslims yak devaranne thoristhiviantha convert madi deshadhrohi madi naraka thorustare muslimarinda doora irabeku”

- **Translation:** “Hindus may believe in God or not, but they are there, but if these Muslims do not show God, they will convert, betray the country and show them hell, one should stay away from Muslims.”

Label: Religion

- **Geo-political:** includes statements that indicate hostility towards geopolitical concerns, relationships with other countries/states, hatred for regional languages, and interstate disputes. These comments are marked with the label “Geo-political”.

Example 6 “Thapevrdalla, bererajya anta gottidru, Nam rajyadallievlnu follow mado lakshagatle janaedaralla??..Nam desha hintour inda uddara adange.”

- **Translation:** “*Its not their mistake, even though they knew it was from a different state, why did so many people in our state follow them?? How will our country progress if it continues like this?*”

Label: Geo-political

- **Violence:** This category includes statements that contain threats, incite violence, or call for violent acts against individuals or groups. Label “Violence” is allocated to comments in this category.

Example 7 “Aa Kitthodhmundege Chappali thagondu hakodubittu yaaru mad-deiro thappu avlenumadilla antha helthyalla Avl Para ne mathadthyalla Thay-ineenu????”

- **Translation:** “*Instead of throwing a slipper at that person, no one is saying that what she did was wrong. She’s just speaking out, isn’t she? Are you her mother????*”

Label: Violence

- **Others:** This category contains comments that convey hatred but do not fit the categories listed above. In our dataset, hate comments against sports, movies, news channels, and comments that do not target any particular group or individual are included in this category. These comments are marked with the label “Others”.

Example 8 “Pavithrage Naresh aasti nodi aase ayta... Naresh gepavithra aunty aasti nodi aase ayta. Fully confused. Upendra song nenpubantu.Aasti nodi aase ayta...!!!!?”

- **Translation:** “*Pavithra got greedy seeing Naresh’s property... Naresh got greedy seeing Pavithraaunty’s property. Totally confused. Got reminded of Upendra’s song. Got greedy seeing the property...!!!!*”

Label: Others

4 Raw data collection

YouTube, a versatile and widely used video-sharing platform, plays a significant role in modern life and society. With its billions of monthly users, it is the world’s largest social media network. YouTube’s diverse array of content, including entertainment, vlogs, music, educational tutorials, news, documentaries, and more, caters to a wide audience. The platform’s global reach and accessibility in multiple languages

make it a hub for intercultural interaction and expression. YouTube also hosts a wide range of content related to political disputes, social commentary, activism, racial issues, community disagreements, and regional variations. The platform's user-generated posts and comments, often in informal and conversational language, reflect contemporary online communication. People from diverse backgrounds use YouTube to express themselves in both their native languages and English, generating rich contextual data crucial for training models that can understand the complexities of hate speech.

We specifically targeted YouTube³ videos on highly relevant and controversial topics such as elections, the change of government in Karnataka, the Hijab issue in Udupi and Mangalore, prominent actors and social media content creators, India-Pakistan border disputes, concerns related to Kashmir, and similar subjects. The comments on these videos, written in English, in Kannada, or in a code-mixed script combining Kannada and English, were then scraped using the YouTube API.

A total of 95,360 comments were scraped for analysis. Since we focus on Kannada-English code-mixed text, we filtered out comments entirely in English or Kannada using the LangDetect library.⁴ However, the results from LangDetect were not entirely accurate, so a manual review of the corpus was necessary to remove irrelevant comments. After this filtering process, we arrived at a final set of 8,058 comments, which were considered for annotation. Extraneous data, such as URLs, symbols, and emoticons, were removed to prepare the data for annotation. We took great care to protect users' privacy, deleting all user profile information, including user IDs and usernames. Additionally, columns such as timestamps, likes, dislikes, reply counts, reply authors, published timestamps, and updated timestamps were excluded. Responses to comments were not deleted but were treated as individual comments.

5 Annotation setup

The extensive use of social media generates a huge amount of data every minute, which is significant for building AI models to automate the identification of hate speech. To ensure these models are trained with high-quality data, data labeling is crucial. The authors have chosen the manual labeling/human annotation method to maintain the annotation quality. This process was carried out in two distinct levels: first, the comments were labeled for binary "Hate" or "Non-hate" labels, and then in the next level, the hate labels were further categorized into more fine-grained classes such as "Gender", "Political", "Religion", "Geo-political", "Violence", and "others", based on their intention towards these categories.

³ <https://www.youtube.com/>.

⁴ <https://pypi.org/project/langdetect/>.

5.1 Manual/human annotation

Human annotation is the process of labeling, reviewing, and classifying data by individuals who understand the context, nuances, and subtleties of the language. This process involves a series of steps to be followed, with a key emphasis on building effective guidelines for annotators. These guidelines are crucial in achieving high-quality data annotation, as they include defined labels, the annotation process, and examples for correct and incorrect labels. The process also involves the selection of annotators with proper backgrounds and language skills, and an evaluation measure to maintain the consistency and reliability of the annotated data.

5.1.1 Annotator selection

To annotate the corpus, 100 students of Visvesvaraya Technological University were considered out of which 75 are undergraduate students and 25 are post-graduation students. Our primary focus was on final-year undergraduate B.Tech and final-semester M.Tech students. An instructional session was conducted to educate students about detecting hate speech on social media and corpus building. A Google form was distributed to the students to express their interest in the annotation process. This helped us to comprehend the students' backgrounds, gender, and language proficiency. Further, a pilot study was conducted to choose the best annotators among the students. In the pilot study, 50 comments were initially labeled for "Hate" and "Non-hate" classes by language experts and the authors, and these were considered the standards for comparison. The same comments were then given to the students for labeling, emphasizing their crucial role in the process. The responses received from the students were compared with the standard comments labeled by the language experts and authors. We retained students whose labels matched at least 60% with the standards. Students who performed poorly were eliminated from the process. The detailed distribution of annotators is depicted in Table 2. It was found that the number of female students exceeded that of male students in both undergraduate and postgraduate studies. Additionally, most students came from urban areas and had completed their schooling in English medium. Despite this, they demonstrated better proficiency in Kannada. The selected students were further trained on annotation guidelines to further label the hate comments into more fine-grained classes.

Table 2 Annotator distribution

Course	Gender	Count	Urban	Rural	Kannada-medium	English-Medium
UG	Female	23	18	5	11	12
	Male	17	13	4	7	10
PG	Female	7	4	3	4	3
	Male	6	3	3	4	2

5.1.2 Annotation guidelines

The authors have established a comprehensive set of guidelines to maintain consistency in the annotation. For hate speech, the guidelines released by the Facebook community for its social media users⁵ were utilized. The annotation process is conducted in two levels, ensuring a thorough categorization. In Level 1, comments are categorized as "Hate" or "Non-hate". In level 2, the hate comments are further categorized into multiple labels namely "Gender", "Political", "Religion", "Geo-political", "Violence", and "Others" based on their orientation towards a particular topic. These guidelines were used to train the students during the pilot study, further reinforcing the comprehensive nature of our approach.

5.1.3 Data labeling

The final dataset, consisting of 8,058 records, was divided into multiple Google Sheets, each containing 500 comments, with columns allocated for three annotators to provide labels. These sheets were distributed among annotators for labeling. To ensure high-quality annotations and avoid overburdening annotators, they were instructed to label 200 comments per day. In the first level of annotation, each annotator independently reviewed each comment and labeled it as "Hate" or "Non-hate" based on the provided guidelines. Three annotators labeled each comment, and the final label was determined using a majority voting approach. In the second level of annotation, comments labeled as hate were further classified into fine-grained categories. In cases of disagreement among annotators, particularly in the fine-grained classifications, the final label was assigned by subject experts to ensure consistency and accuracy. This systematic approach, meticulously designed, was implemented to produce a reliable and robust dataset for subsequent analysis. Table 3 presents sample comments from the annotated corpus; also, the final distribution of labels in the proposed corpus is illustrated in Table 4.

5.1.4 Annotator's agreement: Cohen's Kappa

During annotation, each annotator was instructed to select only one category to which the comment belongs, following the provided guidelines. This collaborative effort, with multiple annotators working on the same dataset, made it necessary to establish a metric for comparing the quality of annotations. This led to the use of the inter-annotator agreement, which measures how well the team conducted annotations on the same dataset. It also indicates the level of agreement among annotators regarding a category, although it does not necessarily confirm the accuracy of the annotations. High inter-annotator agreement suggests that the guidelines are clear and interpretations are consistent, reinforcing our team's collaborative spirit.

To assess the reliability and consistency of hate speech annotations in Kannada-English code-mixed text, Cohen's Kappa score is employed to evaluate inter-annotator

⁵ <https://transparency.fb.com/en-gb/policies/community-standards/hate-speech/>.

Table 3 Samples from the HASTIKA corpus

Comment	Binary class	Multi-class
Evlg e janmukke buddi baralla ansutte.. evldu batte bichiro full video na YouTube ge upload madudre thale yethkondu thirgokagalvalla agadru buddi barutte.. please yaradru first e kelsa maadi Translation: "Looks like this girl won't gain any sense in this lifetime. If someone uploads her nude video on YouTube, she might finally get embarrassed and learn a lesson. Someone please be the first to do this."	Hate	Gender
E sala siddu ge dalit vote beda aste oogiri makke power beku aste aa vayyanige dalitru sattre enu lekka ella Translation: This time Siddhu doesn't need dalit vote, that's all, shame on him, he needs power, that's all, it doesn't matter to him if dalits die	Hate	Politics
Obbha muslim Saha iddanna kandisalla. Ella Muslims support madtare. Ellaru Hindu virodighalu. Translation: Even a single Muslim doesnot raise any objections for this. All Muslims support this. All are anti-Hindu	Hate	Religion
Tamilanadu Janagalu karnataka da yella companies allu edrae e Nan maklu na hodisbeku ha kaade kaveri neer kelthare ekade yella companies kelsa kithkothidare namma Karnataka sarkara kannadigarige meesalithi kottilla, formers ge neeru kooda kodsthilla waste govt. Translation: The people of Tamil Nadu come and work in all the companies in Karnataka, yet they demand water from the Kaveri River. All the companies here are hiring them, and our Karnataka government has not given any reservations or priority to Kannadigas. Even the government is giving away water to these people	Hate	Geo-political
Ee terrorist nann maklu bandu namma desha haal madi ellaranu sayista idare.. Translation: These terrorists have come here, ruined our country, and are killing everyone	Hate	Violence
Loo tirbooke BTv avare yavaloo bethale adavalu kudisikondnu yeno nimdhu thu..loo aha anchor dress nodroo media annutharenoo tukaligala.. Translation: Disgusting BTv people, always showing nude, shameless women. What is this nonsense of yours? Look at the dress of that anchor -do you even call this media? Worthless folks	Hate	Others
Nanu yavattu intaha curiosity match node illa ant anstu.. astondu romanchanakari waw waw Translation: I've never watched such a thrilling match before- it felt so exciting, wow wow!	Non-hate	NA

Table 4 Label distribution of HASTIKA

Label	Binary class		Multi-class					
	Hate	Non-hate	Gender	Political	Religion	Geo political	Violence	Others
Count	3950	4108	1703	699	478	232	277	561

Table 5 Inter annotator agreement for binary class

Pair of Annotators	Hate Label	Non-hate Label
Annotator1 Vs Annotator2	93.4%	96.3%
Annotator2 Vs Annotator3	94.5%	95.5%
Annotator1 Vs Annotator3	96.5%	92.8%
Average	94.8%	94.86%

Table 6 Inter annotator agreement for multi-class

Annotators pair	Gender	Political	Religious	Geo-political	Violence	Others
Annotator1 vs Annotator2	95.3%	93.5%	92.5%	90.5%	92.5%	93%
Annotator2 vs Annotator3	92.6%	94.3%	95.6%	92.3%	93.6%	95.3%
Annotator1 vs Annotator3	94.3%	95.7%	94.6%	91%	95.2%	93.3%
Average	94.06%	94.5%	94.23%	91.26%	93.76%	93.86%

agreement. This quantitative metric measures the level of agreement between annotators, accounting for the possibility of chance agreement. A Cohen's Kappa score of 90% indicates perfect agreement, while a score exceeding 75% reflects substantial agreement beyond chance, and values less than that indicate disagreement and are subject to relabeling. In the annotation process, a high Cohen's Kappa value indicates consistent adherence to the standards by annotators in identifying comments as hate or non-hate and further classifying hate comments into specific categories. A near-perfect Kappa score (above 90%) is achieved for both binary and multi-class labeling, as shown in Tables 5 and 6.

6 Topic modelling

Topic modeling, a widely used statistical method in data analysis, is used to uncover the underlying subtopics, text patterns, and their co-occurrence with other documents in a large corpus. The method revolves around three key elements: 'document', 'word', and 'topic'. Each of these elements plays a crucial role in the process. In our proposed dataset, each comment is considered as a document, represented as a sequence of words, and denoted by a vector of natural numbers.

6.1 Data pre-processing

The preprocessing steps for the proposed HASTIKA corpus are performed to clean and standardize the data for improved performance. First, text normalization is employed whereby all the texts are reduced to lower cases to normalize the text. Noise removal is a crucial step in pre-processing which involves the removal of factors such as URLs, hashtags, mentions, emojis, and other characters that

are non-alphanumeric and do not add much value but rather increase the noise within the dataset. Then tokenization is applied, which splits the given text into its individual tokens, followed by stop word removal. Stop words from both Kannada and English were removed; however, contextual stopword adjustment was crucial as words like ‘Sir’, ‘Devru’(God), ‘Kannadiga’, ‘Namma nadu’(Our Country/State), ‘Avanu’(He), ‘Avalu’(She), ‘Dharma’(Religion), etc. carry specific cultural or contextual meanings within the hate speech are retained as they contribute majorly in identifying the underlying topic as well as classification of hate speech. Further, lemmatization is applied to bring the words to their root form to ensure all their variations are treated as the same word. To enhance the data normalization, repetitive words commonly used in social media for emphasis are reduced to their standard forms. This pre-processed data is used for topic modeling and baseline experiments.

6.2 Latent dirichlet allocation(LDA)

Latent Dirichlet Allocation (LDA) is a robust framework employed for topic modeling and is widely used in unsupervised learning tasks involving large text corpora. Introduced by Blei et al. (2003), LDA is a probabilistic generative model identifying the hidden thematic structure within documents. It represents each document as a probability distribution over topics, and each topic as a distribution over words. The model assumes that documents are mixtures of multiple topics, with words drawn from them according to their respective probabilities. The idea that various documents share the common topics, but each document may exhibit the topics in varying proportion is formalized by the LDA.

Generative process of LDA: Given a collection of documents D , vocabulary of words V , and latent topics K . the generative process can be described as follows:

1. for each topic $k \in \{1, 2, \dots, K\}$, sample a distribution of words $\phi_k \sim \text{Dirichlet}(\beta)$, where β , a hyperparameter that controls the sparsity of the topic-word distribution.
2. for each of the documents d , sample a distribution over topics $\theta_d \sim \text{Dirichlet}(\alpha)$, where α is the hyperparameter that controls the sparsity of the document-topic distribution. For each word W_n in the document, sample a topic Z_n from Multinomial(θ_d) and sample a word w_n from the topic-specific word distribution Multinomial(ϕ_{Z_n}).

Hence each word can be drawn by selecting first a topic and then selecting a word from the corresponding topic distribution. For document d , the joint distribution of the LDA is given by:

$$P(\theta_d, Z_d, W_d \mid \alpha, \beta) = P(\theta_d \mid \alpha) \prod_{n=1}^{N_d} P(Z_{dn} \mid \theta_d) P(W_{dn} \mid Z_{dn}, \beta) \quad (1)$$

To infer the latent variables θ , ϕ , and Z , LDA maximizes the following likelihood function:

$$P(W \mid \alpha, \beta) = \prod_{d=1}^D \int P(\theta_d \mid \alpha) \left(\prod_{n=1}^{N_d} \sum_{Z_{dn}} P(Z_{dn} \mid \theta_d) P(W_{dn} \mid Z_{dn}, \beta) \right) d\theta_d \quad (2)$$

The parameters α and β control the distribution of topics within documents and words within the topics.

In this study, we perform topic modeling on a Kannada-English code-mixed dataset using Latent Dirichlet Allocation (LDA). The pre-processed data is tokenized and fed into the LDA model, where we iteratively refine the number of topics to ensure meaningful representation. The final topics are visually analyzed and labeled for better interpretability.

For implementation, we utilized the NLTK library⁶ for text preprocessing, pandas⁷ for data analysis, seaborn⁸ for visualization, Gensim⁹ for LDA modeling, and pyLDAvis¹⁰ for topic visualization. Further, to ensure topic quality, we evaluated clusters using coherence scores, which measure semantic similarity among key terms. While increasing the number of topics can improve coherence, excessive topics may lead to overlapping and less meaningful clusters. Thus, we balanced interpretability with accuracy by refining topic selection.

Hyperparameter tuning played a crucial role in optimizing the LDA model. The α parameter, set to 'auto' in Gensim, dynamically adjusted the topic distribution across documents, while β was kept at its default value to regulate word distribution within topics. These parameters significantly influence topic sparsity- α controls the number of topics per document, while β affects word allocation per topic.

To determine the optimal number of topics, we employed perplexity measures, UCI coherence scores, and iterative refinements. While perplexity is a standard metric for model evaluation, it proved inadequate due to the complexity of the code-mixed dataset. Instead, we prioritized coherence scores and manual refinements to ensure well-defined and interpretable topic clusters.

Initially, the number of topics for modeling was set to 15, yielding the highest coherence score of 0.44, as depicted in Fig. 2. However, while this setup achieved a high coherence score, the resulting clusters exhibited significant overlap among various topics. The number of topics was reduced to 12 to address this issue, achieving a slightly improved coherence score, as shown in Fig. 3. Nevertheless, the overlapping clusters persisted. To mitigate this, the number of topics was aligned with the predefined labels in the dataset. Despite this adjustment, overlapping topics remained, indicating the need for further refinement. The

⁶ <https://www.nltk.org/>.

⁷ <https://pandas.pydata.org/>.

⁸ <https://seaborn.pydata.org/>.

⁹ <https://pypi.org/project/gensim/>.

¹⁰ <https://pypi.org/project/pyLDAvis/>.

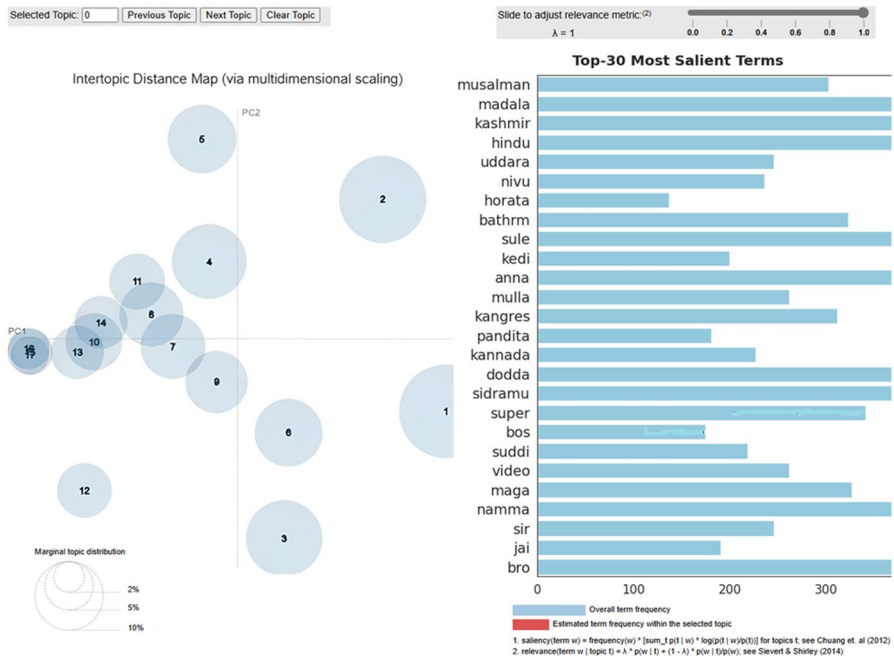


Fig. 2 Topic visualization for a number of topics = 15

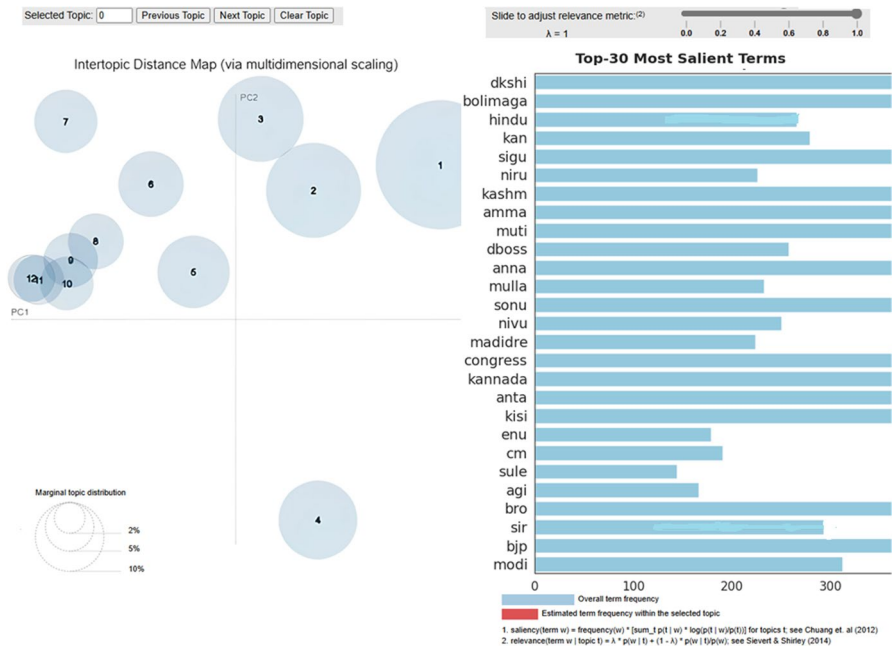


Fig. 3 Topic visualization for a number of topics = 12

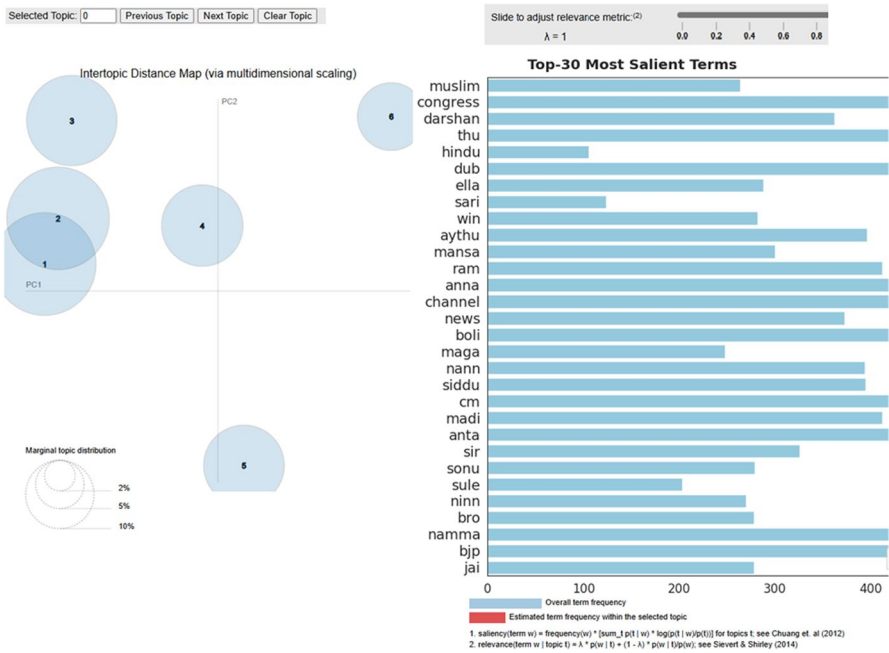


Fig. 4 Topic visualization for a number of topics = 6

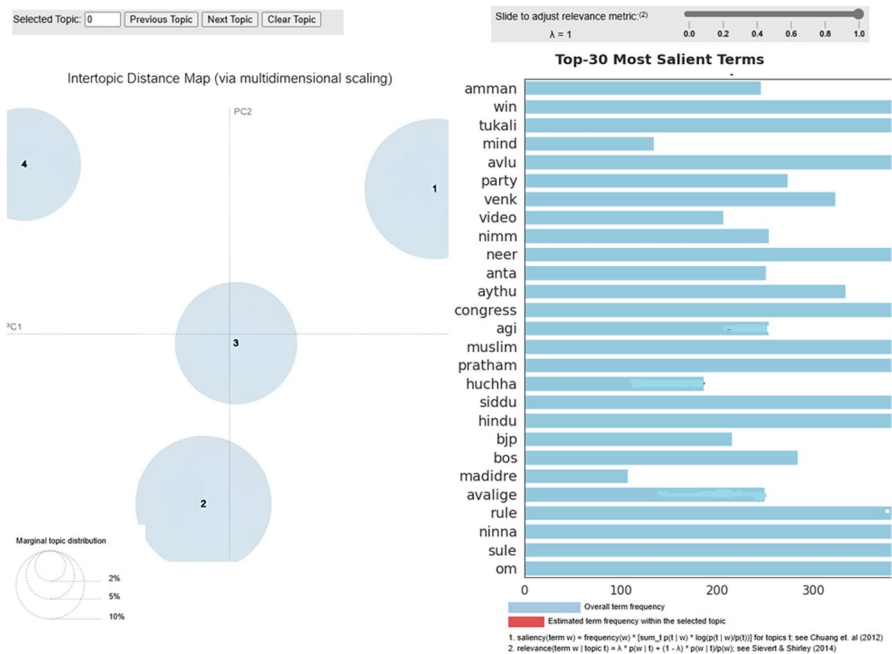


Fig. 5 Topic visualization for a number of topics = 4

results of topic modeling with six topics are presented in Fig. 4. Using an iterative approach, the number of topics was progressively reduced, ultimately settling on four distinct topics. This final configuration produced clear, non-overlapping clusters with a coherence score of 0.34, as shown in Fig. 5. Although the coherence score was marginally lower, the distinct and non-overlapping clusters offered a more interpretable representation of the dataset's underlying themes.

The final topic model identified four distinct clusters, each representing a prevalent and sensitive theme where hate speech frequently occurs. By setting the number of latent topics to four, the model uncovered critical areas that highlight the key societal issues often targeted by harmful and derogatory language. These clusters offer valuable insights into the nature and underlying drivers of hate speech. Each cluster encapsulates the thematic essence of words associated with its respective topic. Table 7 summarizes the topics and the dominant words defining each cluster. This structured representation aids in understanding the linguistic patterns and contextual nuances of hate speech within the dataset.

The key observations from the topic modeling are as follows:

- **Gender Dynamics and Social Issues:** Cluster 1 emerged as the largest and most prominent cluster in the topic modeling analysis. It encompasses terms and phrases associated with gender roles and gender-based discrimination. The cluster is characterized by the presence of salient terms, including slurs, obscene language, and insults specifically targeting individuals or groups based on their gender identity. This cluster reflects the pervasive nature of gender-related hate speech, highlighting the need for focused interventions to address and mitigate such harmful discourse.
- **Political Discourse and Governance:** Cluster 2 primarily revolves around political discourse, with the most common salient terms including references to political parties, figures, elections, policies, and corruption. The language within this cluster often features derogatory expressions, accusations, and inflammatory rhetoric, reflecting the contentious and polarized nature of discussions on political matters.
- **Religious Beliefs and Practices:** Cluster 3 focuses on religion and religious practices, encompassing terms and phrases that often include derogatory and

Table 7 Dominant words in each cluster

Cluster	Topic	Dominant words
1	Gender Dynamics and Social Issues	Obscene language ('sule', 'dagar', 'munde', 'nayi', 'th**a'), name of a celebrity (***), clothing ('batte')
2	Political Discourse and Governance	Political parties ('Congress', 'BJP'), political leaders ('CM', 'siddu', 'modiji'), 'commission', 'guarantees', names of other political members
3	Religious Beliefs and Practices	Religions ('hindu', 'muslim', 'islam', 'dharma', 'hijab') name of the religious leaders
4	Geopolitical Events and Analysis	About the country, states, and languages of India ('desha', 'Tamilnadu', 'Kaveri', 'Kerala')

inflammatory language aimed at specific religious communities and their customs. This cluster highlights the sensitive and contentious nature of discussions surrounding religious topics, where hate speech frequently emerges.

- **Geopolitical Events and Analysis:** Cluster 4 addresses topics related to territorial disputes within Indian states, geographical and political tensions, regional and international conflicts, and linguistic differences. The terms in this cluster often include derogatory language targeting individuals or groups involved in such disputes and conflicts, reflecting the divisive and inflammatory nature of these issues.

In addition to the four primary clusters representing major themes in the dataset, several smaller clusters emerged, defined by their unique sets of terms. These topics were named 'Sports', 'Movies', 'Cauvery River Water Issues', 'Social Media Content Creators', 'TV Channels', and 'Terrorism'. The 'Sports' topic included terms such as 'cricket', 'King Kohli', 'IPL', 'RCB', 'Jadeja', 'Sania Mirza', and 'Tennis'. The 'Movies' topic featured words like 'DBoss', 'Rajakumar', 'Appu', and 'KGF'. The 'Cauvery River Water Issues' topic contained terms such as 'Kaveri', 'Tamil Nadu', 'Karnataka', 'Horata', and 'Neeru'. The 'Social Media Content Creators' topic included words like 's**u', 'YouTube', 'reels', and 'batte'. The 'TV Channels' topic was characterized by terms like 'TV9', 'BTV', 'anchor', and 'news'. Lastly, the 'Terrorism' topic included words such as 'deshadrohi', 'bomb', 'gundu', 'terrorist', and 'kole'.

The visualization of the topic modeling results, presented in Fig. 4, provides a detailed overview of the latent topics within the dataset. The Intertopic Distance Map illustrates the distinct nature of the identified thematic clusters, offering a visual representation of their separation. The marginal topic distribution highlights the uneven prevalence of comments across these topics, emphasizing that certain themes dominate the dataset. The proximity of topics within the map reflects their similarity in terms of shared words and phrases. While the visualization demonstrates that topic modeling effectively captured the dataset's underlying thematic structure, the overlap between clusters 2 and 3 crossing the principal component (PC) lines suggests these topics are not perfectly separable using only the features of PC1 and PC2. This overlap implies that additional latent features and factors influence these clusters, requiring further exploration to achieve clearer differentiation.

7 Exploratory data analysis

The statistical study and exploratory data analysis (EDA) of the data are pivotal for understanding the corpus and evaluating its effectiveness for research. EDA serves as an essential initial step in examining the structure, distribution, and patterns within the data. This analysis helps identify biases, anomalies, and outliers that may influence the results. Moreover, EDA allows for the visualization of correlations between different variables and aids in uncovering the underlying characteristics of the corpus. A thorough understanding of the data is key to building more accurate

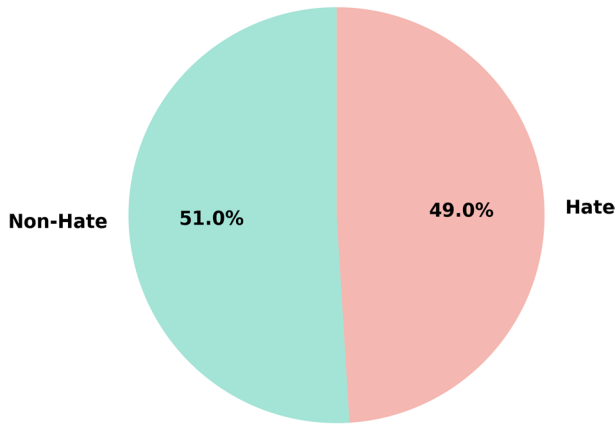


Fig. 6 Binary class label distribution

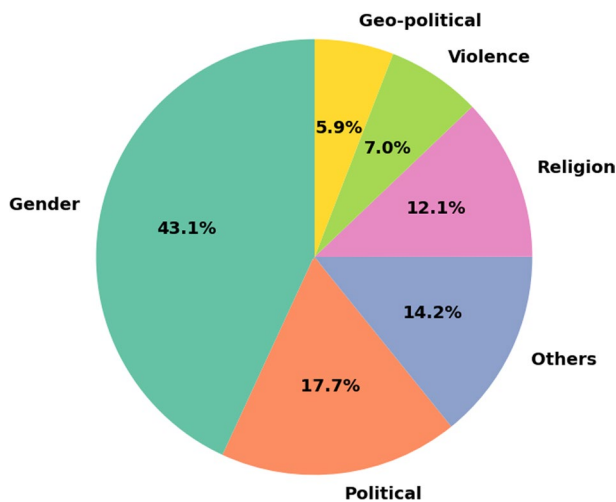


Fig. 7 Multi-class label distribution

and reliable models. In this study, Python libraries such as Matplotlib¹¹ and Seaborn were employed to generate effective and insightful visualizations, providing a deeper understanding of the dataset's attributes.

¹¹ <https://matplotlib.org/>.

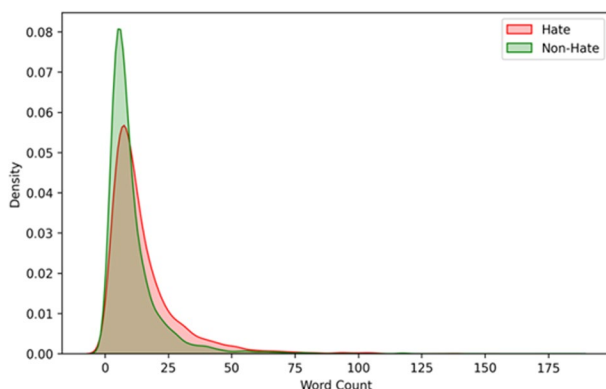
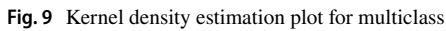


Fig. 8 Kernel density estimation plot for binary class

We began by analyzing the distribution of comments in the dataset based on binary labels: hate and non-hate. This analysis is critical as it directly impacts the balance between the hate and non-hate speech classes, thereby influencing the training and evaluation of machine learning models. Figure 6 illustrates the proportion of each category, offering a clear depiction of the dataset's composition. Comments labeled as "Hate" constitute 49% of the total dataset, while those labeled as "Non-hate" comprise 51%. The minimal difference between these two categories, as shown in Fig. 6, indicates that the corpus is well-balanced for binary classification, without exhibiting any significant bias toward either label. Furthermore, we examined the distribution of comments across the fine-grained hate speech categories. Figure 7 presents the distribution of comments within these categories, which include "Gender", "Religion", "Political", "Geo-political", "Violence", and "Others". The Gender category accounts for the largest proportion, with 43.1%, followed by the Political category at 17.7%. Conversely, the Geo-political category represents the smallest share, at 5.9%. These findings reveal that while the corpus is balanced for binary classification, it exhibits an imbalance in the fine-grained hate categories, with a notable predominance of gender-related hate comments.

Kernel Density Estimation (KDE) plots were generated to analyze the linguistic characteristics and variability in word usage across both binary and multi-class labels. Figure 8 illustrates the KDE plot depicting the distribution of word counts within the binary classes. The plot reveals that the "Non-hate" class distribution is predominantly skewed towards shorter word counts, whereas the "Hate" class exhibits longer sentence lengths in the proposed corpus. Notably, there is an overlap between the two distributions, particularly in the region corresponding to lower word counts. This overlap suggests that some instances in both categories share similar linguistic features, highlighting potential challenges in distinguishing between hate and non-hate comments solely based on sentence length.



Parameter	Binary Class		Multi-class					
	Hate	Non -hate	Gen- der	Poli- tical	Reli- gion	Geo- political	Viole- nce	Others
Avg word count	12.40	9.65	11.48	12.74	17.36	14.88	12.69	11.96
Avg word length	6.64	6.27	6.01	7.87	6.58	6.34	6.27	6.06
Avg no. tokens	10.87	8.76	9.43	14.54	20.49	16.86	14.43	14.39

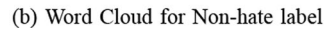
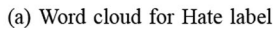


Fig. 10 Word clouds for binary class labels

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Word clouds are another valuable tool for analyzing linguistic corpora, providing a visual representation of word frequency or prominence within the dataset. These visualizations offer an intuitive overview of the most significant terms, reflecting key topics and themes. In this study, word clouds were generated for the binary classification categories, "Hate" and "Non-hate", as illustrated in Fig. 10, and for the fine-grained hate speech categories, including "Gender", "Political", "Religion", "Geo-political", "Violence", and "Others", as depicted in Fig. 11. In these word clouds, the size of each word corresponds to its frequency and prominence within the corpus. Larger words represent dominant topics, while smaller words indicate less frequent and more specific themes. These visualizations effectively highlight linguistic trends and patterns associated with various hate speech categories, thereby enhancing the interpretability and overall analysis of the dataset.

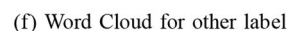
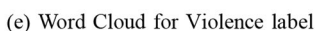
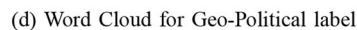
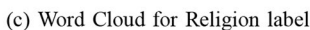
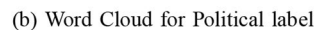
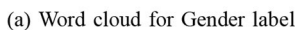


Fig. 11 Word clouds for multi-class labels

8 Baseline methods

This section details the comprehensive methodology employed to evaluate the HASTIKA dataset. Preliminary experiments were conducted to validate the dataset and establish benchmark results. The approach involves systematically evaluating the code-mixed text data using state-of-the-art machine learning and deep learning techniques, alongside advanced methods such as in-context learning. The methodology incorporates word-based feature extraction, sentence-based feature extraction, and prompt-based techniques for in-context learning to perform binary and multi-class classification on the proposed corpus.

8.1 Embedding layer

This subsection outlines the word-based feature extraction techniques utilized in this study, including word2vec, GloVe, and fastText. Furthermore, sentence-level embeddings were obtained using transformer-based models such as BERT, DistilBERT, ALBERT, HateBERT, mBERT, and XLM-RoBERTa. The following subsections provide a detailed explanation of each model and their specific applications to the classification tasks.

8.1.1 word2vec

A semantic-based feature extraction approach, word2vec, is employed to generate dense vector representations for the proposed HASTIKA dataset. This method is used to establish baseline features, with word2vec embeddings of the corpus serving as input for the classification model. The implementation is carried out using the Gensim library, employing word2vec with a vector size of 100, a window size of 5, and a minimum word count of 1, utilizing the skip-gram model for optimal feature generation. library.¹²

8.1.2 GloVe

Global Vectors for Word Representation (GloVe) is a word embedding technique that maps words into continuous vector spaces to establish semantic relationships (Pennington et al., 2014). As a static embedding approach, GloVe generates a single vector representation for each word. It is trained on word co-occurrence statistics from a large corpus, allowing it to represent each word as a dense vector that reflects contextual associations. For the proposed HASTIKA dataset, we employed 100-dimensional vectors to balance the ability to capture complex semantic relationships effectively without excessive computational overhead.

¹² <https://scikit-learn.org/>.

8.1.3 fastText

The fastText embedding method is an extension of the word2vec approach, designed to capture additional contextual information. It is implemented through the Gensim library, with a tokenizer that splits comments into words and subwords, ensuring accurate handling of both Kannada and English. The model is configured with a vector size of 100, a window size of 5, and a minimum word count of 2, focusing on frequent words. fastText utilizes a bag-of-words approach to represent words (Bojanowski et al., 2017), making it particularly useful for morphologically rich languages like Kannada. Additionally, fastText is capable of handling out-of-vocabulary words by generating embeddings for previously unseen words, which is essential for the proposed dataset.

8.1.4 Bidirectional encoder representation(BERT)

BERT (Bidirectional Encoder Representations from Transformers) is a language model developed by Google, recognized for its ability to generate high-quality vector representations. It is trained using the Masked Language Modeling technique in an unsupervised manner. The BERT model integrates both semantic and syntactic features within its embeddings (Devlin, 2018). For the proposed HASTIKA dataset, we employed BERTbase to extract embeddings, which utilizes 12 layers of transformers, a dimensionality of 768, and 12 self-attention heads. BERT also includes special tokens, such as [SEP] (used at the end of a single token) and [CLS] (used as a classification token), as well as the last hidden state of the model, which are crucial for encoding contextual information.

8.1.5 DistilBERT

We employed DistilBERT(Devlin, 2018), a variant of the BERT model that has 40% fewer parameters and 60% faster runtime than the original BERT. DistilBERT is a compact and efficient transformer model trained through knowledge distillation from the BERT base model, retaining 97% of BERT's performance across various downstream tasks. This model was selected for our experiments due to its ability to effectively extract critical features while optimizing resource usage and reducing training time.

8.1.6 ALBERT

As the number of parameters in transformer models increases, the computational power required for calculations also rises. However, this increase in parameters can lead to higher memory usage and potentially reduced precision in performance. To address these challenges, Lan (2019) introduced ALBERT, a lightweight variant of BERT designed for self-supervised language representation learning. ALBERT reduces memory consumption by decreasing the number of parameters through techniques such as cross-layer parameterization, factorized embedding

parameterization, and sentence ordering objectives. Additionally, ALBERT uses a low-dimensional embedding space, which decouples the vocabulary size from the hidden layer size. We selected this model for our experiments due to its ability to efficiently extract critical features while optimizing resource usage and minimizing training time.

8.1.7 HateBERT

HateBERT is a language model derived from retraining the English BERT-base uncased model. It was trained on over one million posts from communities banned on the Reddit platform, specifically tuned to capture language patterns associated with hateful content (Caselli et al., 2021). Given its training on hate-focused data, HateBERT is particularly effective at extracting features relevant to the proposed HASTIKA corpus, enabling it to identify nuanced aspects of hateful language. The vectorized representations generated by HateBERT capture hatefulness across linguistic boundaries by recognizing culturally specific slurs, slang, and subtle indicators of hate speech. As such, HateBERT serves as a valuable tool for detecting hate speech in diverse contexts.

8.1.8 mBERT

Multilingual BERT (mBERT) is a self-supervised model trained on a vast corpus of multilingual data spanning 104 languages (Devlin, 2018). The mBERT model was employed for feature extraction in the proposed dataset due to its robust multilingual capabilities and its ability to understand contextual nuances within text. Given its proficiency in handling multiple languages, mBERT is particularly well-suited for experiments involving code-mixed data. The tokenized text processed by mBERT captures bidirectional contexts, providing rich contextual embeddings for each token, which enhances its performance in tasks involving diverse linguistic structures.

8.1.9 XLM-RoBERTa

XLM-RoBERTa is a multilingual language model trained on 100 different languages using 2.5 TB of Common Crawl data (Conneau et al., 2019). It is inherently capable of handling multiple languages, minimizing the need for extensive pre-processing steps to manage code-mixing. The model's transformer layers capture deep contextual information in both directions, enhancing its understanding of complex linguistic structures. The base version of XLM-RoBERTa consists of 12 hidden layers, 12 attention heads, and 250 K parameters. The high-dimensional vector representations generated by the model serve as feature embeddings for the proposed dataset, enabling effective downstream tasks such as classification and analysis.

8.2 In-context learning

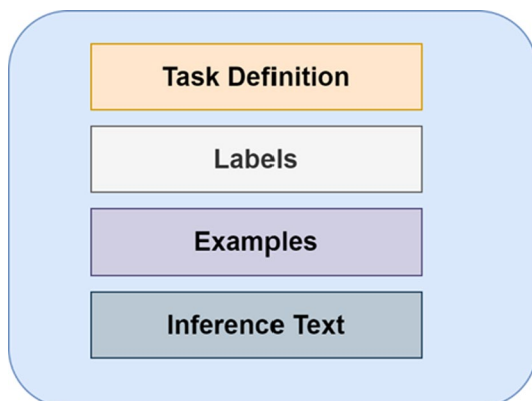
In-context learning (ICL) refers to the ability of a language model to perform a given task by leveraging examples or contextual information provided at the inference stage, without requiring explicit changes to its weights, retraining, or fine-tuning. For instance, text generation is treated as an inference task, with the resulting output referred to as the completion. The entire text or memory available for use within the prompt is referred to as the context window. In-context learning enhances the capabilities of large language models (LLMs) by incorporating examples or secondary information as part of the input. Performance can be further improved by offering relevant examples or context before generating the output or by clearly specifying the task at hand. This paper employs zero-shot, one-shot, and few-shot learning methods to explore the efficacy of ICL. Figure 12 illustrates the overall inference framework.

8.2.1 In-context learning techniques

The general inference template presented in Fig. 12 depicts the structured framework of in-context learning. This template includes a clearly defined task and relevant context, designed to enhance the model's comprehension and performance during inference. The study demonstrates the application of in-context learning for both binary and fine-grained categorization of hate speech using zero-shot, one-shot, and few-shot learning approaches on the proposed dataset.

- Zero-Shot Learning:** Zero-shot learning leverages a model's ability to perform tasks without the need for specific training examples. Instead, the model relies on its pre-trained knowledge to infer the underlying concepts of the task. In this study, zero-shot learning was applied to both binary and fine-grained classification tasks. For binary classification, the model was tasked with distinguishing between "Hate" class and "Non-hate" class based solely on its conceptual understanding of hate speech, acquired during pre-training, as depicted in Fig. 13.

Fig. 12 General template for ICL



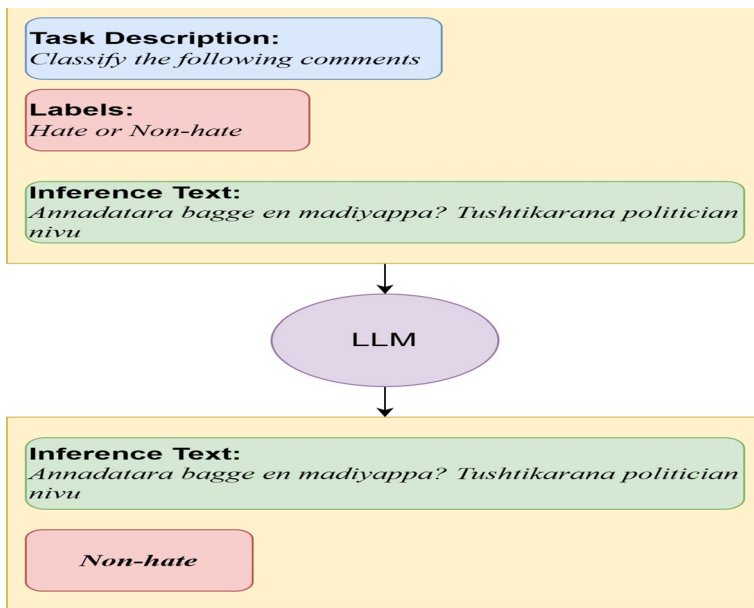


Fig. 13 ICL using Zero-shot prompting

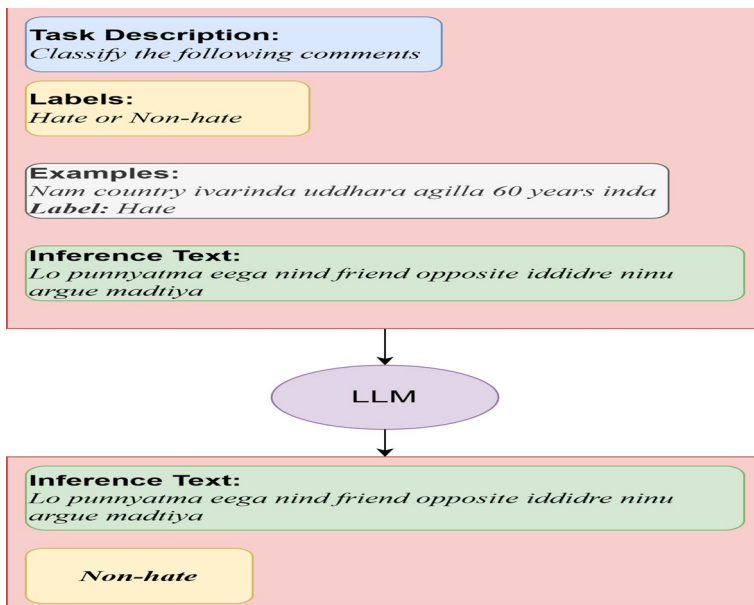


Fig. 14 ICL using One-shot prompting

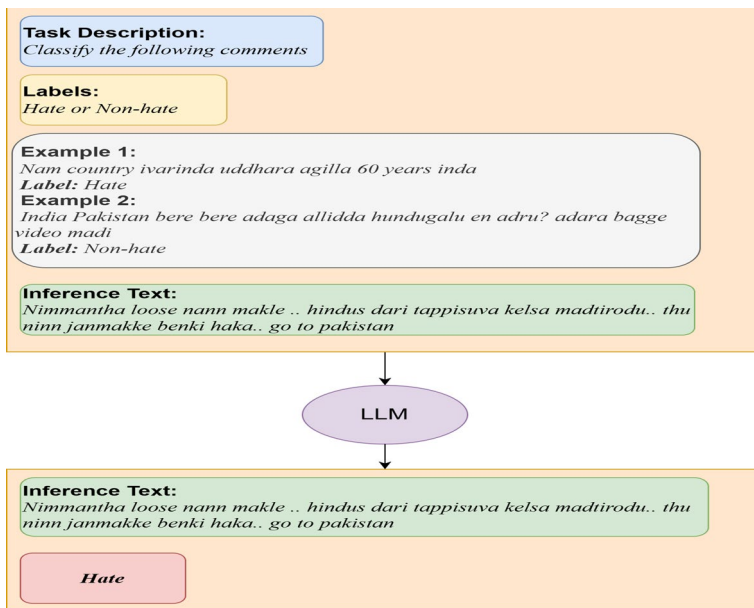


Fig. 15 ICL using Few-shot prompting

Similarly, for fine-grained classification, the model categorized hate speech into the labels “Gender”, “Political”, “Religion”, “Geo-political”, “Violence”, and “Others” without explicit examples. The model’s ability to utilize its learned representations and conceptual understanding enabled it to make informed decisions across these tasks.

- One-Shot Learning:** In this approach, the model is provided with a single example for each class, enabling it to predict the class of a new instance based on the provided example. For binary classification, the model is given one example representing each class, allowing it to grasp and learn the key distinguishing features of the binary classes. Subsequently, the model classifies a new sample into one of the two categories. A visual representation of this process is shown in Fig. 14. Similarly, for fine-grained classification, the model is supplied with a single example for each subclass, facilitating the prediction of the appropriate subclass for a new instance based on the learned features.
- Few-shot Learning:** This approach builds upon the concept of one-shot learning by enabling the model to learn from a small set of examples for each class. For binary classification, a limited number of examples are provided for both the hate and non-hate classes. Similarly, for fine-grained classification, a few representative examples are supplied for each category (e.g., a few examples of hate speech in the gender category, political category, etc.), as depicted in Fig. 15. In the proposed work, five examples were used to facilitate in-context learning. Based on these examples, the model predicts the

Table 9 Performance of word embedding methods for binary classification

Models	Label	F1-score	Accuracy
word2vec+ SVM	Hate	0.7012	0.7097
	Non-Hate	0.6823	
word2vec+ LR	Hate	0.6851	0.6729
	Non-Hate	0.6518	
word2vec+ RF	Hate	0.6452	0.6514
	Non-Hate	0.6511	
word2vec+ KNN	Hate	0.6613	0.6325
	Non-Hate	0.5667	
GloVe+ SVM	Hate	0.7504	0.7337
	Non-Hate	0.7267	
GloVe + LR	Hate	0.7160	0.7081
	Non-Hate	0.7008	
GloVe+ RF	Hate	0.6847	0.6909
	Non-Hate	0.7051	
GloVe+ KNN	Hate	0.6627	0.6431
	Non-Hate	0.6238	
fastText+ SVM	Hate	0.7321	0.7529
	Non-Hate	0.7725	
fastText+ LR	Hate	0.7251	0.7353
	Non-Hate	0.7419	
fastText+ RF	Hate	0.6726	0.6943
	Non-Hate	0.7139	
fastText+ KNN	Hate	0.6837	0.6731
	Non-Hate	0.6414	

category of new samples by leveraging the contextual understanding derived from the provided data.

9 Results and discussion

This section details the results of baseline experiments conducted on the proposed HASTIKA dataset for both binary and multi-class classification tasks. The dataset was partitioned into two subsets, with 80% designated for training and 20% reserved for testing. To ensure fairness and consistency in the evaluation process, a standardized experimental setup was employed across all trials. Following extensive experimentation, the final hyper-parameters were optimized and selected for the experimental trials. Model performance was evaluated using accuracy and F1 score, as these metrics provide a comprehensive assessment of both overall correctness and

Table 10 Performance of transformer models for binary classification

Models	Label	F1-score	Accuracy
BERT	Hate	0.7829	0.8054
	Non-Hate	0.8243	
Distil-BERT	Hate	0.7861	0.7939
	Non-Hate	0.8054	
HateBERT	Hate	0.7739	0.7943
	Non-Hate	0.8109	
ALBERT	Hate	0.7637	0.7738
	Non-Hate	0.8043	
mBERT	Hate	0.7104	0.7519
	Non-Hate	0.7437	
XLM-RoBERTa	Hate	0.7159	0.7657
	Non-Hate	0.7541	

Table 11 Performance of word embedding methods for multi-class classification

Models	F1-score						Acc
	Gender	Political	Religion	Geo-Political	Violence	Others	
word2vec+ SVM	0.7168	0.3729	0.2734	0.1800	0.2218	0.1534	0.5637
word2vec+ LR	0.7016	0.3521	0.2943	0.1729	0.2107	0.1629	0.5627
word2vec+ RF	0.6701	0.3328	0.2648	0.1536	0.1969	0.1607	0.5467
word2vec+ KNN	0.6334	0.3019	0.2515	0.1319	0.2044	0.1431	0.5329
GloVe+ SVM	0.7359	0.4010	0.2955	0.2056	0.2337	0.1757	0.5757
GloVe + LR	0.7210	0.3159	0.3312	0.2453	0.2581	0.1970	0.5600
GloVe+ RF	0.6923	0.2869	0.2967	0.1939	0.2148	0.2109	0.5439
GloVe+ KNN	0.6861	0.2900	0.2841	0.1941	0.2119	0.1923	0.5318
fastText+ SVM	0.7736	0.3858	0.3319	0.2651	0.2548	0.2000	0.6042
fastText+ LR	0.7536	0.3738	0.2841	0.2543	0.2859	0.2105	0.6039
fastText+ RF	0.6851	0.3357	0.3126	0.2357	0.2546	0.1838	0.5651
fastText+ KNN	0.6532	0.3132	0.3010	0.2037	0.1923	0.1556	0.5535

Table 12 Performance of transformer models for multi-class classification

Models	F1-score						Acc
	Gender	Political	Religion	Geo-Political	Violence	Others	
BERT	0.7734	0.7649	0.7637	0.5829	0.3034	0.4518	0.6819
Distil-BERT	0.7736	0.7915	0.7637	0.5243	0.2936	0.4029	0.2367
HateBERT	0.7749	0.7635	0.6459	0.4937	0.3356	0.3314	0.6544
ALBERT	0.7214	0.6923	0.7001	0.5346	0.2819	0.3559	0.6314
mBERT	0.7156	0.7319	0.6818	0.5537	0.3547	0.2943	0.6129
XLM-RoBERTa	0.7071	0.7149	0.5967	0.5619	0.3453	0.3951	0.6139

balance between precision and recall. The experiments were conducted in two stages, focusing separately on binary classification and multi-class classification tasks.

The first subsection presents the results of binary and multi-class classification tasks using both word-based and sentence-based embedding methods. The subsequent subsection discusses the outcomes achieved through in-context learning (ICL) techniques, including zero-shot, one-shot, and few-shot prompting. Finally, a comparative analysis of the best-performing models across word-based, sentence-based, and ICL methods is provided, emphasizing their respective strengths and identifying potential areas for improvement.

9.1 Results achieved through feature learning techniques

In this subsection, we present and analyze the results obtained using word-based and sentence-based feature extraction methods for both binary and multi-class classification tasks.

Table 9 presents the results of binary classification models leveraging word-level embeddings combined with state-of-the-art machine learning classifiers. Table 10 summarizes the performance of transformer-based models for binary classification. Additionally, baseline results for the multi-class classification task are provided in Table 11 and Table 12. From Table 9, it is evident that word2vec combined with an SVM classifier achieves the best performance compared to other machine learning classifiers. However, when paired with classifiers such as Random Forest and KNN, word2vec shows poor performance, particularly for the “Non-hate” class. This observation indicates that distance-based methods like KNN struggle to effectively differentiate between classes. Furthermore, GloVe combined with SVM outperforms word2vec across all metrics, achieving higher F1-scores and accuracy. GloVe’s ability to capture both local and global semantic relationships contributes significantly to its improved performance in classifying complex code-mixed text. Among the word-based embedding methods, the combination of fastText with SVM delivers the highest accuracy and robust F1 scores for both the “Hate” and “Non-hate” classes. fastText’s subword-based representation effectively addresses the complexities of code-mixed text, including out-of-vocabulary words and morphological variations. This capability allows fastText to outperform both word2vec and GloVe, making it particularly well-suited for handling the intricacies of code-mixed text classification.

In addition, we conducted experiments using transformer-based models for sentence-level features, and the results are presented in Table 10. These models leverage pre-trained transformer architectures that encode deep linguistic and contextual information through self-attention mechanisms. Among the models evaluated, BERT achieved the highest performance, with an F1-score of 0.7829 for the “Hate” class and 0.8243 for the “Non-hate” class, resulting in an overall accuracy of 0.8054. The strong F1-score achieved by BERT highlights its ability to balance both classes effectively, minimizing class bias. DistilBERT, a smaller and more efficient transformer model with fewer parameters, delivered performance comparable to

Table 13 Accuracy of LLM models for binary class problem

Models	Zero-shot	One-shot	Few-shot
GPT model	0.6113	0.6761	0.6934
Gemini	0.5845	0.6223	0.6509

Table 14 Accuracy of LLM models for Multi-class problem

Models	Zero-shot	One-shot	Few-shot
GPT model	0.5510	0.5834	0.6155
Gemini	0.5142	0.5520	0.5841

BERT. However, HateBERT, despite being pre-trained on a hateful corpus, did not achieve the same level of effectiveness, indicating limitations in its generalizability to code-mixed text. Furthermore, multilingual models such as mBERT and XLM-RoBERTa demonstrated slightly lower performance, which can be attributed to their cross-lingual training objectives. These objectives, while beneficial for handling multilingual data, may dilute the models' ability to focus on the nuanced characteristics of hate speech in code-mixed text, particularly when the syntax aligns closely with English.

The results for multi-class classification using word-level embeddings are presented in Table 11. Among the word embedding methods, fastText outperformed both word2vec and GloVe across all categories when paired with SVM and Logistic Regression (LR) classifiers. fastText's ability to capture morphological features through subword embeddings likely contributed to its superior performance, particularly in handling the inflected forms prevalent in the code-mixed text. This finding highlights the effectiveness of fastText embeddings, especially when combined with SVM and LR classifiers, in capturing the linguistic complexities inherent in code-mixed text. Among the transformer-based models, both DistilBERT and BERT consistently achieved strong F1-scores in the "Gender", "Political", and "Religion" categories, as shown in Table 12. However, these models exhibited relatively lower F1 scores in the "Geo-political" and "Violence" categories, likely due to the inherent complexity of these topics and the limited number of samples available for training in the dataset. An additional key observation is that cross-lingual and domain-specific pre-trained models did not achieve the same level of accuracy as generalized pre-trained models, indicating that these models may face challenges when applied to specialized datasets with limited domain-specific data.

It is important to note that transformer-based models consistently outperformed traditional word embedding methods such as word2vec, GloVe, and fastText across all categories. This demonstrates that the deep contextual embeddings generated by transformer architectures are more effective in classifying hate speech within a complex, multi-class, code-mixed corpus than traditional embedding techniques. The ability of transformer-based models to capture nuanced contextual information and relationships within the text plays a significant role in their superior performance

Table 15 Comparison of best performing models for binary and multi-class classification

	Feature extraction	Model	Acc
Binary class	Word based	FastText+SVM	0.7529
		BERT	0.8054
		GPT-3.5	0.6113
	Sentence based	Zero-shot	0.6113
		One-shot	0.6761
		Few-shot	0.6934
	Gemini	Zero-shot	0.5845
		One-shot	0.6223
		Few-shot	0.6509
Multi-class	Word based	FastText+SVM	0.6042
		BERT	0.6819
		GPT-3.5	0.5510
	Sentence based	Zero-shot	0.5510
		One-shot	0.5834
		Few-shot	0.6155
	Gemini	Zero-shot	0.5142
		One-shot	0.5520
		Few-shot	0.5841

on tasks involving diverse and intricate linguistic patterns like those found in code-mixed hate speech.

9.2 Results obtained from ICL

The authors conducted a series of experiments using two large language models (LLMs), GPT-3.5 and Gemini, employing the in-context learning (ICL) method to evaluate their performance in identifying hate speech. Three prompting techniques-zero-shot, one-shot, and few-shot-were utilized. For all three cases, the models were prompted using a general template, as shown in Fig. 12. These experiments assessed how effectively the LLMs could classify hate speech based on varying amounts of contextual information provided through the different prompting strategies.

The binary classification results for both models are summarized in Table 13. In the zero-shot setting, the GPT model achieved an accuracy of 0.6113, while Gemini reached 0.5845. With one-shot prompting, both models showed improvements, with accuracies rising to 0.6761 for the GPT model and 0.6223 for Gemini. The few-shot prompting yielded the highest performance, with the GPT model achieving 0.6934 accuracy, while Gemini showed further improvement with 0.6509. Notably, the GPT model consistently outperformed the Gemini model across all three prompting settings. Both models demonstrated enhanced classification accuracy as more context was provided, with the few-shot method yielding the best results.

Table 14 presents the classification report for multi-class classification using in-context learning approaches. In all configurations, GPT-3.5 demonstrated superior performance compared to Gemini, consistent with the results observed in binary classification. This highlights GPT-3.5's stronger contextual understanding and

its greater capacity to generalize as more contextual cues are provided. In contrast, Gemini's classification report indicates that while it performs adequately in binary classification, it faces challenges in more complex multi-class scenarios. Both models showed improved performance as additional context was provided, similar to the findings in binary classification. The inclusion of a few examples for each class was particularly beneficial, aiding the models in capturing the subtleties of fine-grained categories.

Overall, the findings suggest that while few-shot techniques provide a modest improvement to model performance, task-specific fine-tuning with labeled code-mixed text is essential for achieving higher accuracy in hate speech classification. Relying solely on zero-shot or few-shot methods with general-purpose large language models may not be sufficient to capture the complexities of multi-class hate speech detection in low-resource, Kannada-English code-mixed settings, as illustrated in Table 15.

10 Limitations of the proposed work

Although the proposed HASTIKA corpus and methodologies yield promising results in detecting hate speech from Kannada-English code-mixed text, there are several limitations that must be considered. These limitations pertain to the data collection phase, the quality of annotations, the choice of model parameters, and the generalizability of the results beyond the study sample. Acknowledging these limitations is crucial for understanding the scope of the current study and laying the foundation for future efforts to develop more comprehensive and effective hate speech detection systems. Below, we outline some of the key limitations of the present work.

- **Narrow Focus:** The dataset was compiled exclusively from YouTube comments. Other social media platforms, such as Twitter and Facebook, may exhibit different linguistic styles and variations that are not captured in this corpus. As a result, the findings may not fully generalize to all forms of online hate speech across diverse platforms.
- **Disparity of Classes:** The distribution of instances across different classes in the dataset is uneven. Certain classes, such as geo-political and violence, are under-represented, while gender- and political-related hate speech are overrepresented.
- **Bias in Labeling Gender-Related Hate Speech:** Given the female majority in the annotation team, there may be a tendency to over-label certain comments as sexist hate speech due to heightened sensitivity toward derogatory remarks directed at women. While this sensitivity is important, it may also introduce bias, potentially leading to an overrepresentation of gender-related hate speech in the dataset. As a result, the distribution of classes could be skewed, with gender-based hate speech cases appearing more prevalent than they might be in a broader context.
- **Limited Annotator Diversity:** The annotator pool lacked sufficient diversity in terms of age, region, and other socio-cultural factors, which may have influ-

enced the interpretation of comments. Annotators from different age groups, professions, and cultural backgrounds may have varying perspectives on what constitutes hate speech. This limited diversity could introduce biases in the dataset, potentially affecting the representation of hate speech in the proposed corpus.

- **Generalization to Other Dravidian Languages:** The study focuses solely on code-mixed Kannada-English texts and does not encompass code-mixed usage involving other Dravidian languages such as Tamil, Telugu, or Malayalam. These languages may exhibit distinct patterns of code-mixing, which could lead to different linguistic challenges and complexities in hate speech detection.

11 Conclusion and future enhancement

Hate speech detection in code-mixed Dravidian languages, including Kannada-English, presents unique challenges due to varied grammatical structures and frequent language switching. These challenges are further exacerbated by the scarcity of large, well-annotated corpora. To address this gap, the authors introduce HASTIKA, a gold-standard corpus for detecting hate speech in Kannada-English code-mixed text. The dataset has been rigorously annotated for both binary and multi-class classification, ensuring high-quality labels for effective model training. Additionally, the authors provide benchmark results using state-of-the-art machine learning and transformer models for both classification tasks, establishing a strong foundation for future research in this domain. The proposed corpus fills a critical gap in the availability of well-annotated datasets for low-resource Kannada-English settings. Future work should focus on expanding the dataset and fine-tuning classification models to enhance performance, particularly in underrepresented categories. This study lays the groundwork for advancing hate speech detection in Kannada-English code-mixed text, contributing to the larger goal of fostering safer and more inclusive social media platforms.

Appendix

Tables 16, 17, 18, 19 show some of the most salient thematic words from dominant topics.

Table 16 Cluster 1 thematic words (Gender Dynamics and Social Issues)

Words	Translation
Sule	Prostitute
Amma	Mother
Dagar	Prostitute
Shata	Pubic hair
Magane	Son
Bettle	Naked
Avalu	She
Sede	Vulgar
Munde	Widow
Bevarsi	Bastard

Table 17 Cluster 2 thematic words (Political Discourse and Governance)

Words	Translation
Siddu	Name of the politician
Desha	Country
Raita	Farmer
Sulemaklu	Son of prostitute
Modiji	Name of the politician
BJP, congres	Political parties
Kammission	Commission
Sarakara	Government
Gyarantee	Guarantee
Mata	Vote

Table 18 Cluster 3 thematic words (Religious Beliefs and Practices)

Words	Translation
Hindu	Name of the religion
Muslim	Name of the religion
Dharma	Path of rightness
Bhagavatgite	Bhagavad gita
Saadhu	Ascetic
Jihad	Holy war
Hijab	Veil
Islam	Name of the religion
Pandita	Pundit
Virodhi	Hostile

Table 19 Cluster 4 thematic words (Geopolitical Events and Analysis)

Words	Translation
Desha	Country
Tamil	Language
Neeru	Water
Kaveri	Cauvery river
Madras	People from south India
Laude	Obscene word
Pakistan	Name of a country
Kannadigar	Kannadigas
Nayigalu	Dogs
Kalabereke	Conglomeration

Funding No funding was received to assist with the preparation of this manuscript.

Declarations

Conflict of interest No potential Conflict of interest was reported by the authors.

Ethical approval This article does not contain any studies with animals performed by any of the authors. This article does not contain any studies with human participants or animals performed by any of the authors.

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